

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 1 – CONCEPT MAPPING

Course Code	CSA0314	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 1: Concept Mapping
Weightage	35% of CIA	Total Marks	100
CO4 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
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Section/Batch	Slot C /	Date	03-10-2026

Code of Conduct:

I, V.Kailash, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

v.kailash

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts	
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established	
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure	
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence	
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand	

Data Structures for a Weather Monitoring System

PRIMITIVE DATA TYPES

Basic data used to store individual sensor values

Examples:

- float temperature → 23.5°C
- float humidity → 78.2%
- float windSpeed → 12.7 m/s
- int timestamp → 14:30:00
- char unit → °C, %, m/s

NON-PRIMITIVE STRUCTURES

Combine primitive data types to represent complex information

Examples:

- Structure (struct) → temperature, humidity, windSpeed, timestamp
- Array → history[100]
- Record / Tuple → (T, H, W, time)

LINEAR STRUCTURES

Elements are arranged in a sequential order

Examples:

- Queue (FIFO) → real-time sensor readings
- Linked List → ordered weather records
- Stack (LIFO) → recent/temporary readings

NON-LINEAR STRUCTURES

Elements are organized in hierarchical or network form

Examples:

- Tree → organize data by timestamp, location, or sensor type
- Graph → represent relationships between multiple sensors

APPLICATIONS

Used in the actual weather monitoring system

Examples:

- Real-time display of temperature, humidity, and wind speed
- Historical data storage
- Trend analysis
- High-wind alerts
- Multi-sensor monitoring
- Weather forecasting